

# Digital Signal Processing — Complete Worked Solutions Sheet

Step-by-Step Derivations for ALL Numerical Questions Across Past Exam Papers & Lecture Slides | CSE 4631

## 1. FIR FILTER DESIGN & WINDOWED-SINC CALCULATIONS

### Problem 1.1: Standard Low-Pass FIR Filter Design

**Question:** A low-pass FIR filter is to be designed with specifications: passband 10 kHz, stopband 11 kHz, stopband attenuation  $\geq 50$  dB, and sampling frequency  $f_s = 44$  kHz. Superior stopband attenuation is prioritized over faster roll-off. Determine impulse response  $h[n]$ . (Summer 2022-2023 Q1a / Summer 2023-2024 Q4b)

#### Step-by-Step Solution / Derivation:

- Filter Specs: Passband = 10 kHz, Stopband = 11 kHz,  $f_s = 44$  kHz, Attenuation  $\geq 50$  dB.
- Window Selection: High attenuation ( $>50$  dB) requires a Blackman Window (provides  $\sim 74$  dB attenuation,  $\sim 0.02\%$  passband ripple, vs Hamming's  $-53$  dB).
- Normalized Cutoff Frequency ( $f_c$ ): Midpoint between passband and stopband:  

$$f_c = (f_{\text{pass}} + f_{\text{stop}}) / (2 * f_s) = (10,000 + 11,000) / (2 * 44,000) = 21 / 88 = 0.238636$$
- Transition Bandwidth (BW) & Kernel Length (M):  

$$BW = (f_{\text{stop}} - f_{\text{pass}}) / f_s = 1,000 / 44,000 = 1 / 44 \approx 0.022727$$

$$M \approx 4 / BW = 4 / (1/44) = 176 \text{ (even integer; kernel length } M + 1 = 177 \text{ points).}$$
- Complete Filter Kernel Equation:  
 For  $i = 0, 1, \dots, 176$  ( $i \neq 88$ ):  

$$h[i] = K * [\sin(2\pi * 0.238636 * (i - 88)) / (\pi * (i - 88))] * [0.42 - 0.5 \cos(2\pi i / 176) + 0.08 \cos(4\pi i / 176)]$$
 For center sample at  $i = M/2 = 88$  (removable singularity):  

$$h[88] = K * 2\pi * f_c = K * 2\pi(0.238636) = K * 1.4994$$
 where  $K$  normalizes DC unity gain:  $K = 1 / \sum h_{\text{raw}}[j]$ .

### Problem 1.2: Voice Separation Band-Reject Filter Design

**Question:** Design a Blackman windowed-sinc band-reject filter to separate male (85-140 Hz) and female (165-255 Hz) voices from a phone call recorded at  $f_s = 4000$  Hz with transition bandwidth 3 Hz. Calculate  $h(i)$  for  $i$  in  $\{-3, -2, -1, 0, 1, 2\}$  (unshifted/centered indexing). (Summer 2018-2019 Q4a)

#### Step-by-Step Solution / Derivation:

- Filter Specs: Low cutoff  $f_{c1} = 140$  Hz (normalized  $f_{c1} = 140/4000 = 0.035$ ), High cutoff  $f_{c2} = 165$  Hz (normalized  $f_{c2} = 165/4000 = 0.04125$ ). Transition BW = 3 Hz  $\Rightarrow$  normalized BW =  $3/4000 = 0.00075$ .
- Kernel Length (M):  $M \approx 4 / BW_{\text{norm}} = 4 / 0.00075 = 5333.33 \Rightarrow M = 5334$  (even integer). Center index  $M/2 = 2667$ .
- Band-Reject Equation:  $h_{BR}[i] = h_{LP1}[i] + h_{HP2}[i] = h_{LP1}[i] + (\delta[i - M/2] - h_{LP2}[i])$ .
- Evaluation at Center ( $i = M/2$ , or centered index  $k = 0$ ):  
 $h_{LP1}[0] = 2 * f_{c1} = 2 * 0.035 = 0.070$   
 $h_{LP2}[0] = 2 * f_{c2} = 2 * 0.04125 = 0.0825$   
 $h_{BR}[0] = 0.070 + (1.0 - 0.0825) = 0.9875$  (before DC normalization K).
- For adjacent points  $k$  in  $\{-3, -2, -1, 1, 2\}$ :  
 Evaluate sinc<sub>1</sub> - sinc<sub>2</sub> weighted by Blackman window  $w[k + M/2] \approx 0.42 - 0.5 * \cos(2\pi/M) + 0.08 * \cos(4\pi/M) \approx 1.0$  near center.

### Problem 1.3: Calculating Windowed-Sinc Filter Kernel Points

**Question:** Design a Blackman Windowed-Sinc LPF given transition bandwidth BW = 0.4 and cutoff  $f_c = 0.04$  (normalized). Calculate the first four kernel points ( $i = 0, 1, 2, 3$ ) given Blackman window values  $w = (0, 0.05, 0.25, 0.63, 0.95)$ . (Summer 2017-2018 Q3b)

#### Step-by-Step Solution / Derivation:

- Kernel Length:  $M = 4 / BW = 4 / 0.4 = 10$  points. Center  $M/2 = 5$ .
- Formula:  $h_w[i] = 2 * f_c * \text{sinc}(2 * f_c * (i - M/2)) * w[i] = [\sin(2\pi f_c (i - 5)) / (\pi (i - 5))] * w[i]$ .
- Numerical Calculations ( $f_c = 0.04$ ):
  - $i = 0$  ( $w[0] = 0$ ):  $h_w[0] = [\sin(2\pi * 0.04 * (-5)) / (\pi * (-5))] * 0 = 0.0000$
  - $i = 1$  ( $w[1] = 0.05$ ):  $h_w[1] = [\sin(2\pi * 0.04 * (-4)) / (\pi * (-4))] * 0.05 = [\sin(-0.32\pi) / (-4\pi)] * 0.05 = [0.84432 / 12.5664] * 0.05 = 0.00336$
  - $i = 2$  ( $w[2] = 0.25$ ):  $h_w[2] = [\sin(2\pi * 0.04 * (-3)) / (\pi * (-3))] * 0.25 = [\sin(-0.24\pi) / (-3\pi)] * 0.25 = [0.68455 / 9.4248] * 0.25 = 0.01816$
  - $i = 3$  ( $w[3] = 0.63$ ):  $h_w[3] = [\sin(2\pi * 0.04 * (-2)) / (\pi * (-2))] * 0.63 = [\sin(-0.16\pi) / (-2\pi)] * 0.63 = [0.48175 / 6.2832] * 0.63 = 0.04831$

### Problem 1.4: Multi-Stage Cascading for Extreme Stopband Attenuation

**Question:** You need to isolate a 1 nV signal on a 10 V power line (-200 dB relative ratio). You have a single-stage Blackman windowed-sinc filter providing -74 dB stopband attenuation and kernel length 301. How do you design a filter to achieve this, and what is the final stopband attenuation and kernel length? (Summer 2018-2019 Q6b)

#### Step-by-Step Solution / Derivation:

- Design Procedure (Cascading / Self-Convolution): To achieve extreme attenuation beyond a single filter's capability, cascade (convolve) the filter kernel with itself multiple times.
- Number of Stages: Required attenuation =  $20 \log_{10}(1 \text{ nV} / 10 \text{ V}) = -200 \text{ dB}$ .  
For a 3-stage cascade: Total Attenuation =  $3 * (-74 \text{ dB}) = -222 \text{ dB}$  (which exceeds the -200 dB requirement).
- Combined Filter Kernel Length:  
Convolving three 301-point kernels yields length  $L_{\text{total}} = 3 * 301 - 2 = 901$  points.
- Key Precaution: Double-precision floating point (64-bit) must be used during computation to avoid round-off noise floor at -120 dB.

## 2. FILTER KERNEL TRANSFORMATIONS (LP, HP, BP, BR)

### Problem 2.1: Multi-Band Kernel Conversions

**Question:** Given a 7-point low-pass filter kernel  $h_L = \{-1, 0, 1, 2, 1, 0, -1\}$  with cutoff frequency 200 Hz at  $f_s = 1 \text{ kHz}$  ( $f_c = 0.2$ ). Calculate the filter kernels for: i. High-pass filter (cutoff 300 Hz); ii. Band-pass filter (200 Hz to 300 Hz); iii. Band-reject filter (200 Hz to 300 Hz). (Summer 2022-2023 Q5b / Summer 2023-2024 Q3a)

#### Step-by-Step Solution / Derivation:

- i. High-Pass Filter Kernel (300 Hz cutoff):  
Notice cutoff 300 Hz corresponds to  $f_{\{c,HP\}} = 0.5 - f_{\{c,LP\}} = 0.5 - 0.2 = 0.3$  (300 Hz).  
Using Spectral Reversal (modulating by  $(-1)^i$ ):  $h_{HP}[i] = (-1)^i * h_{LP}[i]$ .  
 $h_{HP} = \{(-1)^0(-1), (-1)^1(0), (-1)^2(1), (-1)^3(2), (-1)^4(1), (-1)^5(0), (-1)^6(-1)\} = \{-1, 0, 1, -2, 1, 0, -1\}$ .
- ii. Band-Reject Filter Kernel (200 Hz to 300 Hz):  
Parallel addition of Low-Pass (<200 Hz) and High-Pass (>300 Hz):  
 $h_{BR}[i] = h_{LP}[i] + h_{HP}[i] = \{-1, 0, 1, 2, 1, 0, -1\} + \{-1, 0, 1, -2, 1, 0, -1\} = \{-2, 0, 2, 0, 2, 0, -2\}$ .
- iii. Band-Pass Filter Kernel (200 Hz to 300 Hz):  
Spectral Inversion of Band-Reject (subtracting from unit impulse at center  $i = 3$ ):  
 $h_{BP}[i] = \delta[i - 3] - h_{BR}[i] = \{0, 0, 0, 1, 0, 0, 0\} - \{-2, 0, 2, 0, 2, 0, -2\} = \{2, 0, -2, 1, -2, 0, 2\}$ .

### Problem 2.2: Spectral Inversion vs. Spectral Reversal

**Question:** Convert original low-pass filter kernel  $h = \{-2, -1, 0, 1, 2, 3, 4, 5, 4, 3, 2, 1, 0, -1, -2\}$  to high-pass using: i. Spectral Inversion; ii. Spectral Reversal. (Summer 2018-2019 Q5c)

#### Step-by-Step Solution / Derivation:

- i. Spectral Inversion (LP  $\rightarrow$  HP, same cutoff frequency):  
Formula:  $h_{HP}[n] = \delta[n - M/2] - h_{LP}[n]$ . Center index  $M/2 = 7$  (value = 5).  
Procedure: Negate every sample, then add 1 to center sample at index 7:  
 $h_{HP\_inv} = \{2, 1, 0, -1, -2, -3, -4, -4, -4, -3, -2, -1, 0, 1, 2\}$ .
- ii. Spectral Reversal (LP  $\rightarrow$  HP, shifts cutoff by  $\pi$  / mirrors frequency domain):  
Formula:  $h_{HP}[n] = (-1)^n * h_{LP}[n]$ . Multiply alternating signs (+1, -1, +1, -1, ...):  
 $h_{HP\_rev} = \{-2, 1, 0, -1, 2, -3, 4, 5, 4, -3, 2, -1, 0, 1, -2\}$ .

### Problem 2.3: Frequency Flipping Transformation

**Question:** Transform impulse response  $x[n] = \{1, 3, -4, 7, P, -1, 5, Q\}$  so its corresponding frequency response is flipped left for right. (Summer 2020-2021 Q4c)

#### Step-by-Step Solution / Derivation:

- Procedure: Flipping a frequency response left-for-right (mirroring about  $f_s/4$ ) corresponds to multiplying the time-domain sequence by  $(-1)^n$ .
- Calculation for indices  $n = 0, 1, \dots, 7$ :  
 $x_{flipped}[n] = (-1)^n * x[n] = \{1, -3, -4, -7, P, 1, 5, -Q\}$ .

## 3. DFT ANALYSIS, IDFT SYNTHESIS & COMPLEX OPERATIONS

### Problem 3.1: System Frequency Response in Rectangular Form

**Question:** The rectangular frequency domain input  $X(n)$  and output  $Y(n)$  of a system are given as:  $X = \{80 + 0i, 28 - 14i, -8 + 0i\}$ ,  $Y = \{-80 + 0i, -70 + 140i, 24 + 0i\}$ . Find frequency response  $H(n) = Y(n) / X(n)$  in rectangular form. (Summer 2022-2023 Q3c / Summer 2023-2024 Q2b)

#### Step-by-Step Solution / Derivation:

- Complex Division Formula:  $H[k] = [(ReY * ReX + ImY * ImX) + j(ImY * ReX - ReY * ImX)] / (ReX^2 + ImX^2)$ .
- $k = 0$ :  $H[0] = (-80 + 0i) / (80 + 0i) = -1 + 0i$ .
- $k = 1$ :  $X[1] = 28 - 14i$ ,  $Y[1] = -70 + 140i$ .  
Denominator =  $28^2 + (-14)^2 = 784 + 196 = 980$ .  
Re(Numerator) =  $(-70)(28) + (140)(-14) = -1960 - 1960 = -3920$ .  
Im(Numerator) =  $(140)(28) - (-70)(-14) = 3920 - 980 = 2940$ .  
 $H[1] = (-3920 / 980) + j(2940 / 980) = -4 + 3i$ .
- $k = 2$ :  $H[2] = (24 + 0i) / (-8 + 0i) = -3 + 0i$ .
- Resulting  $H(n) = \{-1 + 0i, -4 + 3i, -3 + 0i\}$ .

### Problem 3.2: Polar Frequency Response to Impulse & Step Response

**Question:** Given system frequency response  $H(n)$  in polar form ( $N = 4$ ):  $\text{MagH} = \{80, 31.305, 8\}$ ,  $\text{PhaseH} = \{0, -0.4636, 3.1416\}$  rad. Find impulse response  $h[n]$  and step response  $s[n]$ . (Summer 2022-2023 Q2a / Summer 2023-2024 Q4c)

#### Step-by-Step Solution / Derivation:

- 1. Convert Polar to Rectangular Form:
  - $\text{ReH}[0] = 80 \cos(0) = 80$ ,  $\text{ImH}[0] = 80 \sin(0) = 0$ .
  - $\text{ReH}[1] = 31.305 \cos(-0.4636) = 28.0$ ,  $\text{ImH}[1] = 31.305 \sin(-0.4636) = -14.0$ .
  - $\text{ReH}[2] = 8 \cos(\pi) = -8$ ,  $\text{ImH}[2] = 8 \sin(\pi) = 0$ .
- 2. Calculate IDFT Synthesis Amplitudes:
  - $\text{ReH\_bar}[0] = \text{ReH}[0]/N = 80/4 = 20$ .
  - $\text{ReH\_bar}[1] = \text{ReH}[1]/(N/2) = 28/2 = 14$ ,  $\text{ImH\_bar}[1] = -\text{ImH}[1]/(N/2) = -(-14)/2 = 7$ .
  - $\text{ReH\_bar}[2] = \text{ReH}[2]/N = -8/4 = -2$ .
- 3. Inverse DFT Synthesis for  $h[n]$  ( $n = 0, 1, 2, 3$ ):
 
$$h[n] = 20 + 14 \cos(\pi n / 2) - 7 \sin(\pi n / 2) - 2 (-1)^n.$$
  - $h[0] = 20 + 14(1) - 7(0) - 2(1) = 32$
  - $h[1] = 20 + 14(0) - 7(1) - 2(-1) = 15$
  - $h[2] = 20 + 14(-1) - 7(0) - 2(1) = 4$
  - $h[3] = 20 + 14(0) - 7(-1) - 2(-1) = 29 \Rightarrow h[n] = \{32, 15, 4, 29\}$ .
- 4. Calculate Step Response  $s[n]$  (Running Sum):
 
$$s[0] = 32; s[1] = 32+15 = 47; s[2] = 47+4 = 51; s[3] = 51+29 = 80 \Rightarrow s[n] = \{32, 47, 51, 80\}.$$

### Problem 3.3: Complete 4-Point Forward DFT Derivation

**Question:** Compute the complete 4-point forward DFT for the time-domain signal  $x[n] = \{1, 2, 3, 4\}$  ( $N = 4$ ). (Lecture Slide Example)

#### Step-by-Step Solution / Derivation:

- Analysis Equations:  $\text{ReX}[k] = \sum x[n] \cos(2\pi kn/N)$ ,  $\text{ImX}[k] = -\sum x[n] \sin(2\pi kn/N)$ .
- $k = 0$  (DC Component):  $\text{ReX}[0] = 1+2+3+4 = 10$ ,  $\text{ImX}[0] = 0$ .
- $k = 1$ :  $\text{ReX}[1] = 1 \cos(0) + 2 \cos(\pi/2) + 3 \cos(\pi) + 4 \cos(3\pi/2) = 1 + 0 - 3 + 0 = -2$ .  
 $\text{ImX}[1] = -[1 \sin(0) + 2 \sin(\pi/2) + 3 \sin(\pi) + 4 \sin(3\pi/2)] = -[0 + 2 + 0 - 4] = 2$ .
- $k = 2$  (Nyquist Component):  $\text{ReX}[2] = 1 - 2 + 3 - 4 = -2$ ,  $\text{ImX}[2] = 0$ .
- $k = 3$  (Conjugate Mirror of  $k = 1$ ):  $\text{ReX}[3] = -2$ ,  $\text{ImX}[3] = -2$ .
- Full DFT Spectrum:  $X[k] = \{10 + 0i, -2 + 2i, -2 + 0i, -2 - 2i\}$ .

### Problem 3.4: Debugging Invalid Real-DFT Implementations

**Question:** Given 16-point real DFT output  $X(n) = \{4+2i, 5-2i, 6+6i, 7-7i, 8+i, 7+i, 6-2i, 5+2i\}$ . Justify why this implementation is incorrect. (Summer 2022-2023 Q6a)

#### Step-by-Step Solution / Derivation:

- Error 1 (DC Bin Rule Violation):  $X[0] = 4 + 2i$  has a non-zero imaginary component ( $\text{Im}X[0] = 2$ ). DC bin must be purely real ( $\text{Im}X[0] = 0$ ).
- Error 2 (Nyquist Bin Rule Violation): For  $N = 16$ , Nyquist bin is at index  $k = 8$ . Either Nyquist bin is missing from array or  $\text{Im}X[8] \neq 0$ .
- Error 3 (Conjugate Odd Symmetry Violation): Real DFT requires  $\text{Im}X[k] = -\text{Im}X[N-k]$ . Here  $\text{Im}X[1] = -2i$  but  $\text{Im}X[15]$  (last element)  $= +2i$  fails symmetry check.

## 4. SIGNAL DECOMPOSITIONS & SYSTEM PROPERTIES

### Problem 4.1: Even-Odd & Interlaced Signal Decomposition

**Question:** Resolve the signal  $x[n] = \{-1, 3, 4, -2, 5, 7, -9, 1\}$  ( $n = 0..7$ ) into: i. Even-odd decomposition; ii. Interlaced decomposition. (Summer 2021-2022 Q2a)

#### Step-by-Step Solution / Derivation:

- i. Even-Odd Decomposition:

Even part  $x_e[n] = (x[n] + x[-n]) / 2$ , Odd part  $x_o[n] = (x[n] - x[-n]) / 2$ .

For  $x[n]$  defined over  $n = 0..7$ ,  $x[-n]$  extends over  $n = -7..0$ .

$x_e[0] = x[0] = -1$ ,  $x_e[n] = 0.5 * x[n]$  for  $n \neq 0$ .

- ii. Interlaced Decomposition (Even vs. Odd sample indices):

Even-indexed sequence:  $x_{\text{even}}[n] = \{x[0], x[2], x[4], x[6]\} = \{-1, 4, 5, -9\}$ .

Odd-indexed sequence:  $x_{\text{odd}}[n] = \{x[1], x[3], x[5], x[7]\} = \{3, -2, 7, 1\}$ .

### Problem 4.2: System Linearity & Superposition Proofs

**Question:** Test linearity for input-output systems: i.  $y(t) = t^2 * x(t-1)$ ; ii.  $y[n] = 2x[n] + 3$ ; iii.  $y[n] = x[n+1] - x[n-1]$ . (Summer 2018-2019 Q2a / Formula Sheet)

#### Step-by-Step Solution / Derivation:

- Superposition Test:  $T\{a*x_1 + b*x_2\}$  must equal  $a*T\{x_1\} + b*T\{x_2\}$ .
- i.  $y(t) = t^2 * x(t-1)$ :  $T\{a*x_1 + b*x_2\} = t^2 * (a*x_1(t-1) + b*x_2(t-1)) = a*t^2*x_1(t-1) + b*t^2*x_2(t-1) \Rightarrow \text{LINEAR}$ .
- ii.  $y[n] = 2x[n] + 3$ : Combined  $= 2(a*x_1 + b*x_2) + 3$ ; Weighted sum  $= a(2x_1+3) + b(2x_2+3) = 2a*x_1 + 2b*x_2 + 3a + 3b$ . Unequal due to offset +3  $\Rightarrow \text{NON-LINEAR}$ .
- iii.  $y[n] = x[n+1] - x[n-1]$ :  $T\{a*x_1 + b*x_2\} = (a*x_1[n+1] + b*x_2[n+1]) - (a*x_1[n-1] + b*x_2[n-1]) = a*y_1[n] + b*y_2[n] \Rightarrow \text{LINEAR}$ .

**Problem 4.3: Energy and Power of the Unit Step Sequence**

**Question:** Determine the energy and power of the unit step sequence  $u[n]$ . (Summer 2023-2024 Q3b)

**Step-by-Step Solution / Derivation:**

- Energy (E):  $E = \sum |u[n]|^2$  from  $n = -\infty$  to  $+\infty = \sum (1)^2$  from  $n = 0$  to  $+\infty = \infty$  (Infinite Energy).
- Power (P):  $P = \lim_{N \rightarrow \infty} (1 / (2N+1)) * \sum |u[n]|^2$  from  $n = -N$  to  $N$ .  
 $P = \lim_{N \rightarrow \infty} (1 / (2N+1)) * (N + 1) = \lim_{N \rightarrow \infty} (N + 1) / (2N + 1) = 1/2 = 0.5 \text{ W}$  (Finite Power).

**5. DISCRETE CONVOLUTION & LTI SYSTEM BLOCK DIAGRAMS****Problem 5.1: Discrete Convolution of Piecewise Sequences**

**Question:** Compute convolution of  $x(n) = u(n+1) - u(n-4) - \delta(n-5)$  with  $h(n) = [u(n+2) - u(n-3)] * (3 - |n|)$ . (Summer 2022-2023 Q1b)

**Step-by-Step Solution / Derivation:**

- Input Sequence  $x[n]$ :  $u(n+1) - u(n-4)$  gives 1 for  $n$  in  $\{-1, 0, 1, 2, 3\}$ . Subtracting  $\delta(n-5)$  adds -1 at  $n = 5$ .  
 $x[n] = \{1, 1, 1, 1, 1, 0, -1\}$  defined over  $n = -1..5$ .
- Impulse Response  $h[n]$ :  $u(n+2) - u(n-3)$  limits  $n$  to  $\{-2, -1, 0, 1, 2\}$ . Evaluating  $(3 - |n|)$ :  
 $h[-2] = 1, h[-1] = 2, h[0] = 3, h[1] = 2, h[2] = 1 \Rightarrow h[n] = \{1, 2, 3, 2, 1\}$  defined over  $n = -2..2$ .
- Discrete Convolution Sum:  $y[n] = x[n] * h[n]$ . Range of  $y[n]$  is  $n = (-1 + -2)$  to  $(5 + 2) \Rightarrow n = -3..7$  (11 points).  
 $y[-3] = 1*1 = 1; y[-2] = 1*2 + 1*1 = 3; y[-1] = 1*3 + 1*2 + 1*1 = 6; y[0] = 1*2 + 1*3 + 1*2 + 1*1 = 8;$   
 $y[1] = 1*1 + 1*2 + 1*3 + 1*2 + 1*1 = 9; y[2] = 9; y[3] = 8; y[4] = 6; y[5] = 2; y[6] = -2; y[7] = -1.$

**Problem 5.2: Parallel & Cascaded LTI System Combination**

**Question:** System  $\tau$  has parallel filters  $h_1 = \{5, -2, 3, 0\}$  and  $h_2 = \{-4, 1, 2, -1\}$  added together and cascaded into  $h_3 = \{2, -1, 1, 3\}$ . i. Find overall  $h[n]$ ; ii. Find output  $y[n]$  for  $x[n] = \{4, -3, 0, 2, 5\}$ . (Summer 2023-2024 Q6c / Summer 2021-2022 Q2b)

**Step-by-Step Solution / Derivation:**

- i. Parallel Combination:  $h_{12}[n] = h_1[n] + h_2[n] = \{5-4, -2+1, 3+2, 0-1\} = \{1, -1, 5, -1\}$ .
- Cascaded Combination:  $h[n] = h_{12}[n] * h_3[n] = \{1, -1, 5, -1\} * \{2, -1, 1, 3\}$ :  
 $h[0] = 2; h[1] = -1-2 = -3; h[2] = 1+1+10 = 12; h[3] = 3-1-5-2 = -5;$   
 $h[4] = -3+5+1 = 3; h[5] = 15+1 = 16; h[6] = -3 \Rightarrow h[n] = \{2, -3, 12, -5, 3, 16, -3\}.$
- ii. Output  $y[n] = x[n] * h[n]$ : Convoluting 5-point  $x[n]$  with 7-point  $h[n]$  yields 11-point output  $y[n]$ .

**6. SAMPLING, QUANTIZATION & DECIBEL CALCULATIONS**

### Problem 6.1: Sampling, Aliasing & SQNR Analysis

**Question:** Analog signal  $x_a(t) = 4 \cos(2000\pi t) + 7 \cos(6000\pi t)$  sampled at  $f_s = 5000$  Hz, quantized at 5 bits/sample. i. Nyquist rate and folding frequency; ii. Can signal be faithfully reconstructed?; iii. SQNR in dB. (Summer 2021-2022 Q1c / Summer 2023-2024 Q6b)

#### Step-by-Step Solution / Derivation:

- i. Frequencies:  $f_1 = 1000$  Hz,  $f_2 = 3000$  Hz  $\Rightarrow f_{\max} = 3000$  Hz.  
Nyquist Rate =  $2 * f_{\max} = 6000$  Hz. Folding Frequency =  $f_s / 2 = 2500$  Hz.
- ii. Reconstruction Check: Since  $f_s = 5000$  Hz  $<$  Nyquist Rate (6000 Hz), aliasing occurs! The 3000 Hz tone aliases to  $|3000 - 5000| = 2000$  Hz. Reconstructed signal is corrupted and CANNOT be faithfully restored.
- iii. SQNR Calculation for  $b = 5$  bits:  
 $SQNR \approx 6.02 * b + 1.76$  dB =  $6.02(5) + 1.76 = 30.10 + 1.76 = 31.86$  dB.

### Problem 6.2: Amplifier Power Ratios in Decibels

**Question:** You want an 800-watt amplifier but can only afford a 350-watt amplifier. How many dB of sound power do you sacrifice? (Summer 2018-2019 Q4c)

#### Step-by-Step Solution / Derivation:

- Power Ratio Formula in Decibels:  $\Delta dB = 10 * \log_{10}(P_2 / P_1)$ .
- Calculation:  $\Delta dB = 10 * \log_{10}(350 / 800) = 10 * \log_{10}(0.4375) = 10 * (-0.3589) = -3.59$  dB.
- Result: You sacrifice approximately 3.59 dB of sound power.

## 7. SUPPLEMENTARY THEORETICAL & MATHEMATICAL MODELS (SLIDES)

### Concept 7.1: Segment Averaging & Noise Reduction Law

**Question:** Explain Segment Averaging (Welch's Method) for DFT noise reduction and its mathematical equivalence to bin averaging. (Chapter 9.pdf)

#### Step-by-Step Solution / Derivation:

- Problem: Longer DFTs increase frequency resolution but do NOT reduce spectral noise level.
- Segment Averaging Law: Break long input into  $N_{\text{segments}}$ , window each, compute DFTs, average magnitude spectra:  
Noise Amplitude  $\propto 1 / \sqrt{N_{\text{segments}}}$ . Averaging 100 segments reduces noise by a factor of 10.
- Mathematical Equivalence: Averaging 64 adjacent bins from a 16,384-point DFT is mathematically equivalent to averaging 64 segments of 256 points each.

## Concept 7.2: Spectral Density & Frequency Bin Bandwidth

**Question:** Why do Inverse DFT scaling factors differ between interior bins ( $2/N$ ) and boundary bins ( $1/N$ )? (DSP\_Smith.pdf / Fourier\_Transform.pdf)

### Step-by-Step Solution / Derivation:

- Spectral Density Concept: Frequency domain values represent spectral density (amplitude per unit bandwidth).
- Bandwidth Allocation: Interior frequency bins  $k = 1..N/2-1$  represent a bandwidth of  $2/N$  (fraction of total sampling rate). Boundary bins at DC ( $k=0$ ) and Nyquist ( $k=N/2$ ) cover only HALF this bandwidth ( $1/N$ ).
- Complex Folding: Interior bins appear as conjugate pairs ( $k$  and  $N-k$ ) in complex DFT; folding them doubles amplitude ( $2/N$ ), whereas DC and Nyquist have no mirror counterpart ( $1/N$ ).

## Concept 7.3: Fourier Family Classification Matrix

**Question:** Outline the Fourier Family Classification Matrix and the Duality Golden Rule. (Fourier Transforms.pdf)

### Step-by-Step Solution / Derivation:

- Fourier Family 4-Quadrant Matrix:
  1. Continuous + Aperiodic  $\Rightarrow$  Fourier Transform (FT) [Continuous & Aperiodic spectrum]
  2. Continuous + Periodic  $\Rightarrow$  Fourier Series (FS) [Discrete & Aperiodic spectrum]
  3. Discrete + Aperiodic  $\Rightarrow$  Discrete-Time FT (DTFT) [Continuous & Periodic spectrum]
  4. Discrete + Periodic  $\Rightarrow$  Discrete Fourier Transform (DFT) [Discrete & Periodic spectrum]
- Duality Golden Rule: Discrete in one domain  $\Leftrightarrow$  Periodic in the other domain.

## Concept 7.4: The 7 Polar Nuisances & Solutions

**Question:** List the 7 Polar Nuisances in DSP and their mathematical solutions. (DSP\_Smith.pdf)

### Step-by-Step Solution / Derivation:

1. Radians vs. Degrees: Trig functions require radians; define  $PI = 3.14159265$ .
2. Divide-by-Zero: Occurs when  $ReX = 0$  in  $\arctan(ImX/ReX)$ ; test  $ReX = 0$  and replace with  $1e-20$ .
3. Arctan Quadrant Ambiguity: Standard arctan returns  $[-\pi/2, +\pi/2]$ ; when  $ReX < 0$ , add or subtract  $\pi$  ( $180^\circ$ ).
4. Phase Instability: Near zero magnitude, round-off noise causes wild phase swings  $[-\pi, +\pi]$ .
5.  $2\pi$  Phase Ambiguity: Discontinuities crossing  $\pm\pi$ ; solved by Phase Unwrapping algorithm (adding  $\pm 2\pi \cdot k$ ).
6. Magnitude Positivity ( $\pi$  Ambiguity): Positive magnitude definition forces  $\pi$  phase jumps when real part dips below 0.
7. Boundary Spikes: Round-off noise near  $\pm\pi$  causes rapid toggling between  $+\pi$  and  $-\pi$ .

**Concept 7.5: Decibel Mathematics & Reference Units**

**Question:** State the fundamental formulas for Decibels and standard reference units. (Digital Filters.pdf)

**Step-by-Step Solution / Derivation:**

- Power Ratio:  $\text{dB} = 10 * \log_{10}(P_2 / P_1) = 4.342945 * \ln(P_2 / P_1)$ .
- Amplitude Ratio:  $\text{dB} = 20 * \log_{10}(A_2 / A_1) = 8.685890 * \ln(A_2 / A_1)$ .
- Key Benchmark Points: -3 dB => Amplitude drops to 0.707 (Power drops to 0.5 / 50%).
- Standard Reference Units: dBV (referenced to 1 V rms); dBm (referenced to 1 mW into 600  $\Omega$  load  $\approx$  0.78 V rms).

**Concept 7.6: Amplitude Modulation (AM) Spectrum Shift Math**

**Question:** Derive the frequency-domain spectrum shift equation for Amplitude Modulation (AM). (DSP\_Formula\_Sheet.pdf)

**Step-by-Step Solution / Derivation:**

- Time-Domain Multiplication:  $y[n] = x[n] * \cos(\omega_c n)$ .
- Frequency-Domain Convolution:  $Y(\omega) = 0.5 * X(\omega - \omega_c) + 0.5 * X(\omega + \omega_c)$ .
- Physical Effect: Time-domain carrier multiplication shifts baseband spectrum  $X(\omega)$  into two symmetric sidebands centered at  $+\omega_c$  and  $-\omega_c$ .